Local Pieces, Global Solutions: A Machine-Checked Sheaf-Cohomological Account of Set-Theoretic Estimation

From Combettes' feasibility sets and Carlson's cryptanalytic STE to a mechanized gluing theorem,
with an empirical corpus instantiation — written to be readable without prior cohomology

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Abstract

Set-theoretic estimation (STE) answers an estimation problem by intersecting property sets: an estimate is acceptable exactly when it is consistent with every piece of information [16]. When the hypothesis space is a product over N variables, the feasibility set lives behind an exponential wall, and the standard escape is *locality*: keep small tables over small, overlapping sets of variables. Every local scheme lives or dies on one question — *when does locally consistent partial information determine a global solution?* The mathematics of that question is sheaf theory and Čech cohomology [3, 4], and this paper develops it for STE in machine-checked form (Lean 4 / Mathlib [17, 34]) while assuming no topology of the reader: every term — cover, section, compatible family, gluing, obstruction — is defined from STE primitives, with its constraint-satisfaction meaning stated in one sentence beside it. The main theorem, mechanized sorry-free in the module `Ste.SupportCover`, is the *support–cover gluing theorem*: over any cover of the variables, every compatible family of local sections glues to a global solution *iff* the constraint is generated by pieces whose scopes fit inside the cover — so gluing decides representability over given scopes, and the entire computational difficulty of STE relocates into *computing local sections*, with gluing itself free. Around this centerpiece we prove exact obstruction counts on coupling constraints, rectangle-cover and fooling-set bounds with machine-checked incomparability results, and a reading of the bounded-treewidth algorithmic stack (bucket elimination, junction trees, adaptive consistency) as the theory of when local sections are cheap. We connect the picture to Carlson's cryptanalytic instantiation of STE [12], whose mechanized counting and reduction theorems we reinterpret sheaf-wise — honestly separating what maps, what is mechanized where, and what is speculation. Finally we report a constructive proof-of-work: a verified solver, run on three historical text corpora, whose per-variable consistency scan is proven equivalent to genuine feasibility — and whose completeness is itself a mechanized theorem: frame-structured corpora are *rectangular*, with zero gluing obstruction over every cover (`Ste.FrameRectangular`), so the linear-time local check is the complete gluing test for this class, and the same theory says exactly where it would stop working. Two further mechanized structures answer, metric-freely, what a feasible family of subsets has that an infeasible one lacks: frame corpora have Helly number 2 — pairwise consistency already certifies global consistency, provably false for general subset families (`Ste.HellyConsistency`) — and the forced-coreference reading of a corpus carries a submodular partition-lattice rank — Shannon's lattice of information with counting rank in place of entropy (`Ste.PartitionRank`). Throughout, a strict bright line separates machine-checked results (each cites its Lean identifier) from conjecture, and every heavy cochain construction is quarantined in a clearly marked appendix.

1 Introduction

This paper tells one story three times, at increasing precision: in plain constraint-satisfaction language, in the vocabulary of sheaves built up from STE primitives, and in the named theorems of a Lean 4 mechanization. The story: an STE problem intersects property sets; intersections over product spaces hide an exponential wall; local reasoning over small variable scopes is the escape; local reasoning is sound and complete exactly when local pieces *glue*; and the gluing question has an exact, machine-checked answer — it succeeds precisely when the constraint is cut out by pieces that fit the chosen scopes. Everything else in the paper — obstruction counts, representation bounds, the algorithmic stack, the cryptanalytic and corpus instantiations — is that one theorem seen from different distances.

Contributions.

1. An *accessible on-ramp* to the sheaf view of STE: every topological word is defined from STE primitives before use (Sections 6–7), with worked examples small enough to check by hand.

2. The *support–cover gluing theorem* (Section 8), mechanized sorry-free (`Ste.SupportCover`): the Čech obstruction of a cover vanishes iff the constraint admits a generating family refining the cover. Corollaries: gluing decides representability over given scopes; the obstruction is a certificate of non-decomposability (`exists_scope_refiningNone_of_not_cechVanishesCover`); and a mechanized "gluing width" invariant (`GluesAtWidth`, in `Ste.GluingWidth`) whose separation from treewidth is now machine-checked on *both* sides: allEqual has gluing width 1 (`allEqual_gluesAtWidth_one`) and treewidth $n - 1$ (`treewidth_primalGraph_allEqualInstance`, in `Ste.CliqueTreewidth`).

3. Machine-checked *quantitative* results (Section 8): exact obstruction counts on coupling constraints, rectangle-cover and fooling-set bounds, and incomparability theorems refuting the slogan "obstruction size = representation blow-up."

4. An honest *Carlson connection* (Section 9): which parts of the dissertation's STE cryptanalysis [12] map onto the sheaf picture — with mechanized witnesses (`Ste.Carlson.SheafBridge`), including the theorem that substitution-cipher keys are genuinely *coupled* (`bijKeys_not_rectangular`), the opposite end of the obstruction spectrum from the corpora of Section 10 — and which parts do not map.

5. An *empirical instantiation* (Section 10): a verified solver run on three historical corpora, whose per-variable check is proven equivalent to genuine feasibility (`consistent_iff_forall_disagreementDegree_le_one`) — and the mechanized cohomological reason it is complete: frame corpora are rectangular, with zero gluing obstruction over every cover (`frame_cechVanishesCover`, in `Ste.FrameRectangular`).

6. Two *metric-free discriminators of feasibility* (Section 11), answering what a feasible structure of subsets has that an infeasible one lacks without any metric: frame corpora have Helly number 2 — pairwise consistency certifies global consistency (`frame_hasHelly_two`), provably false for general subset families (`exists_not_hasHelly_two`), with the property sets generating a mechanized closure system (`cl`, `isClClosed_iInter`) — and the forced-coreference reading of a corpus is a lattice meet (`mustSetoid_eq_sInf`) carrying a submodular partition-lattice rank (`rank_submodular`), the counting shadow of Shannon's information lattice.

The bright line. One convention is enforced throughout and is non-negotiable. Every machine-checked claim cites its Lean identifier in `this font`; the named theorem exists, sorry-free, in the project library under continuous integration. Everything not so cited is explicitly labelled: *conjecture*, *sketch*, *dispatched* (an exact statement handed to an in-flight mechanization effort, not yet part of the trusted base), or *speculation*. Several results were dispatched to mechanization during the writing of this paper; all have since landed, sorry-free, in the main library, and are cited as ordinary theorems — the provenance record is kept in Section 12. Adding intuition never relaxes this line; intuition sits beside the citations, not in place of them.

How to read this paper. A reader who knows STE or constraint satisfaction but no algebraic topology can read straight through: Sections 2–4 need no new vocabulary, Section 6 builds the five words the rest uses, and no commutative diagram appears outside the appendix. Asides marked [**For the cohomologist**] and Appendix A carry the cochain-level machinery and can be skipped without losing the thread.

2 Set-theoretic estimation: foundations

STE, as founded by Combettes [16] (with set-membership antecedents in 32), replaces optimization by *feasibility*. An estimation problem is specified by a solution space Ξ, an index set I of pieces of information, and for each $i \in I$ a *property set* $S_i \subseteq \Xi$: the estimates consistent with that piece of information. The answer object is the *feasibility set*

$$T \;=\; \bigcap_{i\in I} S_i,$$

and a set-theoretic estimate is any point of T. The mechanization of this core is `Ste.Basic`: `feasibilitySet`, `mem_feasibilitySet`, the refinement law `feasibilitySet_subset`, partial enforcement `partialFeasibilitySet` with its monotonicity `partialFeasibilitySet_antitone` (more information never enlarges the feasible set), and Combettes' taxonomy: a formulation is *consistent* if $T \neq \emptyset$, *fair* if the true object lies in T (`Fair`, whence consistency: `feasibilitySet_nonempty_of_fair`), and *ideal* if T is exactly the true object (`Ideal`, `Ideal.eq_of_mem`).

Throughout this paper the hypothesis space is a product: $\Xi = \prod_{v\in V} A_v$, a hypothesis assigning a value to each of N variables, each property set constraining some of them. This is the constraint-satisfaction shape of STE, and it hides a wall. The feasible set lives in a space of size exponential in N, and the project has machine-checked that no representation cleverness can help in general: crisp property sets realize *any* of the 2^N extensional states (`card_allExactStates`, `arbitrary_subset_as_single_property`, in `Ste.DynamicFrame.FiniteSolver`). Any solver that manipulates T extensionally pays 2^N. The rest of the paper is about the principled way around: reason *locally*, and understand exactly when local reasoning is complete.

3 Carlson and the topological turn

Between Combettes' analytic setting (metric and Hilbert spaces, projections onto convex sets) and this paper's combinatorial one sits Carlson's dissertation [12], which recast STE over finite, discrete *topological* spaces and applied it to cryptanalysis. The move matters for us twice over: it is the project's motivating instantiation of STE outside signal processing, and it is the first place in this lineage where the *topological* character of feasibility — closed sets, compactness, intersection structure — is taken seriously as the load-bearing mathematics. This paper's sheaf layer is the continuation of exactly that turn: from the topology of the feasible set to the topology of *where information lives*.

3.1 Decryption as STE

Carlson's recast, mechanized in `Ste.Carlson.Cipher`: the solution space is a finite key space; each observed ciphertext c contributes the property set of keys under which c decrypts to a meaningful message; decryption succeeds when the feasibility set collapses to the true key — Shannon's unicity point [33] in set-theoretic form. Machine-checked: the cipher-system skeleton (`CipherSystem`, unambiguous decryption `decrypt_unique`), the induced property sets (`keyPropertySet`) and feasible keys (`feasibleKeys`, `mem_feasibleKeys`), fairness of genuine traffic — if the observations really are encryptions under a true key, that key is feasible (`fair_of_genuine`, `feasibleKeys_nonempty_of_genuine`) — unicity as ideality (`Unicity`, `Unicity.eq_of_feasible`), and observation monotonicity: more intercepted traffic can only narrow the key search (`feasibleKeys_antitone`, a special case of `partialFeasibilitySet_antitone`).

The topological well-posedness facts Carlson's setting needs are mechanized in `Ste.Topology`: closed property sets give a closed feasibility set (`isClosed_feasibilitySet`; discrete case `isClosed_feasibilitySet_of_discrete`), fairness gives the finite intersection property (`FiniteConsistency.of_fair`), and compactness turns finite consistency into consistency outright (`feasibilitySet_nonempty_of_compact`); ideal formulations have closed feasible sets in any T_1 space (`Ideal.isClosed_feasibilitySet`).

3.2 Counting and reduction, machine-checked

Two quantitative pillars of the dissertation are mechanized exactly.

Residual key counting (`Ste.Carlson.Counting`). For a substitution cipher over an alphabet A — keys are permutations of A — an intercepted message constrains the key only on the set T of symbols that actually occur. Carlson's Lemma 4.1 counts the keys equivalent to the true key: exactly $(|A| - |T|)!$ of them (`card_consistent_keys`, via `card_perm_fixing`; the key space itself has size $|A|!$, `card_substitution_keys`). This is the size of the STE feasible key set after one observation — the residual ambiguity.

Cipher reduction (`Ste.Carlson.Reduction`). On n-bit blocks, every permutation-cipher key (a permutation of bit positions) is realized as a substitution-cipher key (a permutation of the value space) by the induced map `coperm` (`permutation_reduces_to_substitution`), injectively (`coperm_injective`); but for $n \geq 2$ there are strictly more substitution keys than permutation keys (`card_permutation_lt_card_substitution`), so no faithful embedding is onto (`substitution_not_reducible_to_permutation`) — Carlson's Lemma 5.1/Corollary 5.2 — and permutation–substitution–permutation compositions collapse to a single substitution key (`psp_reduces_to_substitution`, Corollary 5.3; the same reduction is developed in 13, 25).

Section 9 returns to Carlson with the sheaf vocabulary in hand and asks, honestly, which of these constructions are secretly about local sections and gluing.

4 Why sheaves and cohomology are a useful play

The escape from the 2^N wall that every practical system reaches for is *locality*: instead of the monolithic T, keep small tables about small, overlapping sets of variables — per-variable domains, arc-consistent pair tables, junction-tree bags — and reason with those. Every such scheme lives or dies on one question:

> *If I know everything the problem permits locally — on each small set of variables — and my local answers never contradict each other, do they determine a genuine global solution?*

Sometimes yes, sometimes no. When yes, local reasoning is *complete*: the cheap, small tables are a faithful proxy for the exponentially large T. When no, the local tables admit *phantoms* — mutually consistent local answers realized by no global solution — and any margins-only procedure wastes work on them, or worse, reports them. The mathematical bookkeeping for exactly this local-versus-global question is called sheaf theory, and the device that counts the failures is called Čech cohomology. The names do not matter; the payoff does:

- **An exact criterion** for when local reasoning over given scopes is complete (the support–cover theorem, Section 8) — machine-checked, with no structural side conditions on how the scopes overlap.
- **Certificates**: a single stuck family of local answers certifies that *no* decomposition of the constraint respects those scopes; a fooling set certifies that no small union-of-rectangles representation exists (Section 8.5).
- **A precise location for hardness**: the theory proves gluing is never the expensive step — computing genuine local tables is — which turns the bounded-width algorithmic stack into a completeness theorem for local reasoning (Section 8.4).
- **A no-free-lunch theorem**: any invariant computed from the shape of the scopes alone, ignoring the tables, is provably useless (machine-checked; Appendix A.1) — which tells you where *not* to look for speedups.

Where can the local-to-global question even fail? The development contains three distinct axes of locality, and it is worth fixing at the outset that the question is only genuinely hard on one of them.

1. **Splitting the constraint list** (index set I). Partition or arbitrarily cover the constraints, intersect within groups, intersect the results: you always recover exactly T. The map $J \mapsto \mathrm{feas}(J) = \bigcap_{i \in J} S_i$ sends unions of constraint sets to intersections of partial feasible sets (`partialFeasibilitySet_iUnion`), so any cover of the constraint index recovers the global feasible set (`feasibilitySet_eq_iInter_cover`, in `Ste.Sheaf`). *Local-to-global holds here for free*; nothing can obstruct it. (Under the formal-concept reading of `Ste.HyperFrame`, feas is a lower polar of the satisfaction context [24] (`partialFeasibilitySet_eq_lowerPolar`), and polars always turn joins into meets.)
2. **Splitting the variable set** (scopes $W \subseteq V$). Keep, for each small scope, the partial assignments the problem permits. Here local-to-global genuinely fails, the failure is precisely variable *coupling*, and this is where the paper lives.
3. **Splitting the evidence corpus** (document sets D, in `Ste.DynamicFrame`). Feasibility over $D \cup E$ splits exactly: feasible on the union iff feasible on each piece *and* the finite, explicitly computable set of constraints active only on the union is satisfied

(`crossObstruction`, `mem_feasible_union_iff`, in `Ste.DynamicFrame.PresheafGluing`); a common extension of compatible local worlds exists iff that correction set is empty, and is then unique (`globalExtension_nonempty_iff`, `globalExtension_subsingleton`, `glue`). A local-to-global theorem *with an exact error term*, already closed.

The slogan the mechanization supports: *the constraint side glues for free, the corpus side glues with an exact and computable correction term, and the variable side is where the real local-versus-global phenomenon occurs.* From here on, "local" means local in the variables.

5 Related work

Sheaf-theoretic contextuality. The program closest to this paper is Abramsky–Brandenburger's sheaf-theoretic treatment of non-locality and contextuality [3], its cohomological refinement [4], the logical reading via paradox [5], contextual semantics across disciplines [2], the quantitative contextual fraction [6], and the studies of when cohomological invariants are complete or have false negatives [1, 14, 15, 30]. The present work differs in two ways: the section presheaf of STE consists of *globally extendable* partial assignments (their empirical models are arbitrary compatible data), which collapses the per-section obstruction (`amb_extension_always`, Appendix A.2) and moves the interesting content to family-level gluing; and every result here claimed as proven is machine-checked in Lean 4 [17, 34].

Structural constraint satisfaction. The algorithmic layer this paper reinterprets is classical: consistency enforcement [23, 29], constraint networks and directional/adaptive consistency [20], tree clustering [21], bucket elimination [18, 19], with the width theory of Bertelè and Brioschi [9], Robertson and Seymour [31], Arnborg et al. [8], Bodlaender [10], and the homomorphism-side tractability characterization of Grohe [26]. The paper's contribution to this literature is not a new algorithm but a mechanized exact boundary: which covers admit complete local reasoning at all, independent of acyclicity hypotheses.

Representation lower bounds. The rectangle-cover number and fooling sets are imported from communication complexity [7, 28]; the extension to LP extension complexity [22, 35] is flagged as outlook.

STE lineage. Foundations [16], set-membership estimation [32], and the cryptanalytic line [12, 13, 25, 33].

6 Background: cohomology, built from STE primitives

This section defines every piece of vocabulary the paper uses. Each definition is an ordinary constraint-satisfaction object; the topological name is recorded so that the Lean files and the literature can be read, but nothing later depends on knowing why topologists chose it.

6.1 The five words

Scope, and cover. A *scope* (the Lean files say *context*) is a subset $W \subseteq V$ of the variables. A *cover* is a choice of scopes $U : J \to \mathcal{P}(V)$ such that every variable appears in at least one ($\forall v\, \exists j,\ v \in U_j$). That is all a cover is: **a choice of overlapping variable-scopes to reason locally over**.

Three examples recur: the *singleton cover* $\{\{v\} : v \in V\}$ (pure per-variable reasoning); the *bag cover* given by the bags of a junction tree or elimination order; the *trivial cover* $\{V\}$ (no locality at all).

Local section. For a constraint $T \subseteq \prod_v A_v$ and a scope W, a *local section of T over W* is **a partial assignment on W that extends to a full solution of T** — the restriction to W of some member of T. The set of them, $\mathcal{S}_T(W) = \texttt{localSections}\, T\, W$, is the image of T under coordinate restriction (`Ste.VariablePresheaf`). Note the convention carefully: a local section is not merely "locally plausible"; global extendability is baked into the definition. (Partial assignments that are only locally consistent — what a propagation algorithm computes — are a weaker object; Section 8.4 names them *pseudo-sections*, and the distinction carries much of the story.) Restriction behaves as it should — restricting in two steps equals restricting in one (`restrict_restrict`, `localSections_restrict`). The topologist's word for "a scope-indexed family of section sets with well-behaved restriction" is a *presheaf*; that bookkeeping is all the word means here.

Compatible family. Fix a cover U. A *compatible family* is a choice of one local section $s_j \in \mathcal{S}_T(U_j)$ per scope such that any two agree on the overlap of their scopes (`CompatibleFamily`, in `Ste.CechCover`): **local answers that never contradict each other on shared variables**. This is exactly the state of knowledge of a solver that has computed perfect local tables and checked them pairwise.

Gluing. A compatible family *glues* (`GluesCover`) when a single global solution $f \in T$ restricts to s_j on each U_j: **the local pieces can be stitched into one global solution**. When every compatible family over U glues, the *sheaf condition* holds for U (`CechVanishesCover`); in topological notation "$\check{H}^1$ vanishes for the cover," and its STE meaning is exactly **"every locally consistent choice is globally realizable."**

Obstruction. When the sheaf condition fails, the failure can be counted:

$$\text{obs} \;=\; \#\{\text{compatible families}\} \;-\; \#\{\text{families that glue}\},$$

the number of ways local answers can be mutually consistent yet stitchable into no global solution. The singleton-cover count is `cechObstruction`; the cover-level count is `coverCechObstruction`, and it vanishes whenever the sheaf condition holds (`coverCechObstruction_eq_zero`). Each stuck family is a *phantom*: a profile of partial assignments that every margins-only procedure must entertain although no solution produces it.

Table 1 collects the dictionary.

Why would a compatible family ever fail to glue? Each piece individually extends to a global solution — that is the definition of a local section. The catch: each piece may extend to a *different* solution, with no single solution extending all of them at once. That one sentence is the entire phenomenon; the next subsection exhibits it in four rows of a truth table.

6.2 The smallest example, worked by hand

All statements in this subsection and the next are proved in Lean, sorry-free (the singleton-cover computations live in `Ste.CechObstruction`); theorem names are cited exactly.

topological word	STE meaning	Lean name
context / open set	scope: a subset $W \subseteq V$ of variables	—
cover	choice of overlapping scopes hitting every variable	(hypothesis of `CechVanishesCover`)
local section over W	partial assignment on W extending to a solution	`localSections`
presheaf	scope-indexed section sets, consistent restriction	`restrict_restrict`
compatible family	local answers agreeing on shared variables	`CompatibleFamily`
gluing	stitching the pieces into one global solution	`GluesCover`
sheaf condition ($\check{H}^1 = 0$)	every compatible family glues	`CechVanishesCover`
Čech obstruction	count of compatible-but-unstitchable families	`cechObstruction`, `coverCechObstruction`

Table 1: The five-word dictionary. Everything in this paper is one of these objects.

Two Booleans that must agree. Take two Boolean variables x, y and the single constraint "$x = y$":

$$\Delta \; = \; \{f : \{x, y\} \to \mathsf{Bool} \mid f(x) = f(y)\}$$

(`diagonal`, in `Ste.Sheaf`). It has exactly two solutions, $(\mathsf{ff}, \mathsf{ff})$ and $(\mathsf{tt}, \mathsf{tt})$ (`diagonal_encard`). Reason locally over the singleton cover: one scope $\{x\}$, one scope $\{y\}$.

Local sections. Over $\{x\}$: does $x = \mathsf{ff}$ extend to a solution? Yes ($(\mathsf{ff}, \mathsf{ff})$). Does $x = \mathsf{tt}$? Yes ($(\mathsf{tt}, \mathsf{tt})$). So both values of x are local sections, and symmetrically for y (`contextSections_diagonal`): every partial view is fully uninformative — each variable, seen alone, can still be anything.

Compatible families. The two scopes do not overlap (`singletonCover_overlap_empty`), so there is nothing to agree on: compatibility is automatic, and a compatible family is just a choice of one value per variable — four families in all (`compatibleFamilies`, `compatibleFamilies_diagonal_encard`):

$$(\mathsf{ff}, \mathsf{ff}) \quad (\mathsf{ff}, \mathsf{tt}) \quad (\mathsf{tt}, \mathsf{ff}) \quad (\mathsf{tt}, \mathsf{tt}).$$

Which glue? A family glues iff, read as a full assignment, it is a solution (`glues_iff_mem`, `gluedFamilies_eq`). Two do (`gluedFamilies_diagonal_encard`). The mixed family $x \mapsto \mathsf{ff}$, $y \mapsto \mathsf{tt}$ is the interesting one: **each half extends to a solution, but to different solutions**, and no single solution extends both. It is compatible and stuck, machine-checked as `diagonal_mixed_compatible_not_glues`. The obstruction count is

$$\mathrm{obs}(\Delta) \; = \; 4 - 2 \; = \; 2$$

(`diagonal_cechObstruction`). Two phantoms: the two mixed families. This is what a nonzero Čech obstruction *is* — nothing more exotic than the mixed rows of this truth table.

The exact boundary: gluing iff no coupling. Call T *rectangular* if it is a product of per-variable sets, $T = \prod_v P_v$ (`IsRectangle`) — the constraint imposes no coupling, only per-variable filters. Rectangular problems are the fragment where per-variable reasoning is complete: the feasible set of a rectangular family is itself a rectangle (`rectangular_feasibilitySet`) of cardinality $\prod_v |\bigcap_i P_{i,v}|$ (`rectangular_encard`), representable in $\sum_v$ space, and its singleton-cover obstruction is zero (`rectangular_cechObstruction`). The diagonal is provably not rectangular (`diagonal_not_rectangular`); worse, the smallest rectangle containing it is the whole four-point space (`rectangle_superset_diagonal_encard`) — the tightest per-variable view of a coupling carries *zero* information. The singleton-cover story is closed as an iff (`cechVanishes_iff_rectangular`, in `Ste.CouplingLowerBound`):

$$\text{every compatible family glues} \iff T \text{ is a rectangle.}$$

In plain terms: **per-variable local reasoning is exactly as strong as the absence of coupling**. At this cover the obstruction does not merely notice coupling; it characterizes it.

Scaling: n variables that must all agree. The phantom count is not an artifact of small examples. The n-fold coupling $\mathrm{allEqual}(n, \alpha)$ — all n coordinates equal, over an alphabet α (`allEqual`) — has every per-variable margin full (`contextSections_allEqual`), exactly as the diagonal did: locally, anything goes. So *all* $|\alpha|^n$ families are compatible (`compatibleFamilies_allEqual_encard`), while only the $|\alpha|$ constant assignments are solutions (`allEqual_encard`). The obstruction is exactly $|\alpha|^n - |\alpha|$ (`cechObstruction_allEqual`), hence at least $2^n - 2$ once $|\alpha| \geq 2$ (`cechObstruction_allEqual_ge`): exponentially many phantoms while the true feasible set keeps constant size. For $n \geq 2$ the constraint is genuinely non-rectangular (`allEqual_not_rectangular`, `allEqual_not_cechVanishes`). A margins-only pipeline on this instance does exponential work distinguishing phantoms from solutions — the 2^N wall of Section 2, seen from the local side.

7 The dictionary, precisely

For reference, this section restates the correspondence at full precision; it adds no new ideas beyond Section 6.

Definition 1 (The section presheaf; mechanized, `Ste.VariablePresheaf`). *For $T \subseteq \prod_{v \in V} A_v$ and $W \subseteq V$, the local sections are $\mathcal{S}_T(W) = \{\, f|_W \mid f \in T \,\}$ (`localSections`, the image of T under $f \mapsto (v : W) \mapsto f\, v$). For $W' \subseteq W$ the restriction map $\mathcal{S}_T(W) \to \mathcal{S}_T(W')$ is coordinate restriction; functoriality is `restrict_restrict` and `localSections_restrict`. This makes $W \mapsto \mathcal{S}_T(W)$ a contravariant functor on scopes — a presheaf of sets.*

Definition 2 (Compatibility, gluing, vanishing; mechanized, `Ste.CechCover`). *For a cover $U : J \to \mathcal{P}(V)$: a family $(s_j)_{j \in J}$ with $s_j \in \mathcal{S}_T(U_j)$ is* compatible *(`CompatibleFamily`) if s_j and s_k agree on $U_j \cap U_k$ for all j, k; it* glues *(`GluesCover`) if some $f \in T$ has $f|_{U_j} = s_j$ for all j; the constraint satisfies the sheaf condition for U (`CechVanishesCover`) if every compatible family glues. Glues along a cover are unique (`glueCover_unique`: two solutions agreeing on every scope of a cover agree everywhere), so the sheaf condition is genuinely existence-and-uniqueness.*

Four machine-checked anchor facts frame the landscape:

- *Rectangles glue over every cover*, uniquely (`rectangular_cechVanishesCover`, `rectangular_glueCover_existsUnique`).
- *The trivial cover glues everything* (`cechVanishesCover_univ`): with one scope containing all variables, "local" already means "global."
- *The singleton cover recovers the closed theory* of Section 6.2 (`cechVanishesCover_singleton_iff`), and the diagonal witnesses genuine failure at the cover level (`diagonal_not_cechVanishesCover`).

- *The obstruction count refines the condition* (`coverCechObstruction`, `coverCechObstruction_eq_zero`; the tautological box-gap form at the singleton cover is `cechObstruction_eq_box_sub_encard`).

[For the cohomologist.] The correspondence with the standard apparatus: scopes are the opens of the "space" V (discrete), $\mathcal{S}_T$ is a subpresheaf of the product-of-values presheaf, compatibility is the Čech 0-cocycle condition for the cover, gluing is exactness of $F(V) \to \check{H}^0(U, F)$, and the obstruction count is a cardinality surrogate for the cokernel. The linearized versions — honest cochain complexes with constant and presheaf coefficients, the degree-one quotients, and the comparison with Abramsky–Mansfield–Barbosa's γ — are mechanized and discussed in Appendix A.

So gluing depends on the cover, monotonely in spirit: coarse covers glue easily, fine covers glue rarely, and somewhere in between sits a boundary. The next section states the machine-checked theorem that locates the boundary exactly.

8 Main theorems: gluing works exactly when the pieces fit the cover

8.1 The support–cover gluing theorem

One more everyday notion. A constraint S has *support* $\sigma \subseteq V$ if membership in S depends only on the coordinates in σ (`HasSupport`, from `Ste.Support`) — σ is the constraint's *scope* in the ordinary CSP sense. Say a family (S_i) with scopes σ_i *generates* T if $T = \bigcap_i S_i$, and say its scopes *refine* the cover U if every σ_i fits inside some single scope U_j.

In plain language first: **stitching local answers into global solutions works, for every compatible family, exactly when the constraint can be written as a conjunction of pieces each of which fits inside one scope of the cover.** If the cover's scopes are big enough to each swallow whole constraints of some generating family, gluing is guaranteed; if no generating family fits, some compatible family is stuck. There is no third case.

Theorem 1 (Support–cover gluing; **mechanized**, `Ste.SupportCover`). *Let $T \subseteq \prod_v A_v$ and let $U : J \to \mathcal{P}(V)$ be a cover of the variable set.*

(a) If T is generated by a family whose scopes refine U, then every compatible family of local sections glues to a global solution — `CechVanishesCover` $T\,U$ holds (`cechVanishesCover_of_refiningGenerators`).

(b) Conversely, if `CechVanishesCover` $T\,U$ holds, then $T = \bigcap_j \pi_{U_j}^{-1}\big(\mathcal{S}_T(U_j)\big)$: T is generated by the cylinder *constraints "your restriction to U_j is a local section" (`sectionCylinder`, supported on their scopes by `hasSupport_sectionCylinder`) — a generating family whose scopes are exactly the scopes of U; indeed gluing is* equivalent *to this cylinder identity (`cechVanishesCover_iff_eq_iInter_sectionCylinder`).*

Hence the Čech obstruction of the cover U vanishes iff *T admits a generating family whose scopes refine U — packaged as "gluing $\iff$ there is a per-scope generating family S_j with `HasSupport` $(S_j)\,(U_j)$" (`cechVanishesCover_iff_exists_scopedGeneration`).*

Proof idea (exposition only; the machine-checked proof is in `Ste.SupportCover`). (a) Given a compatible family (s_j), build the only candidate global assignment there is: for each variable v, pick any scope $U_{j(v)}$ containing v and set $f(v) := s_{j(v)}(v)$. Agreement on overlaps makes the choice of scope immaterial, so f restricts to s_j on *every* scope. Why is f a solution? Check the generators one at a time. A generator S_i has its whole scope σ_i inside some single U_{j_0}; on that scope, f agrees with s_{j_0}, which agrees with the global witness $f_{j_0} \in T$ that makes s_{j_0} a local section. Since S_i only reads coordinates in σ_i, and f matches a known solution there, $f \in S_i$. Every generator passes, so $f \in T$. *The single place the hypothesis is used is that each generator's scope fits inside one scope of the cover* — a constraint straddling two scopes could read a mixture of two different witnesses and fail. (Uniqueness of the glue is `glueCover_unique`.)

(b) The cylinders visibly contain T; conversely an assignment f in all cylinders restricts to a local section on every scope, those restrictions automatically agree on overlaps (they all come from f), gluing produces a solution g agreeing with f on every scope, and since the scopes cover V, $g = f$ (again `glueCover_unique`), so $f \in T$. □

Remark 1 (Sanity anchors, all machine-checked)**.** *At the singleton cover, "a generating family with singleton scopes" means "a rectangle," and direction (b) specializes to* `cechVanishes_iff_rectangular`*. Direction (a) contains* `rectangular_cechVanishesCover`*: singleton scopes refine* every *cover. The diagonal's failure is consistent: its minimal scope is the full pair $\{x, y\}$ (*`diagonal_support_full`*, in* `Ste.Support`*), contained in no singleton. The trivial-cover success (*`cechVanishesCover_univ`*) is the case where the single scope swallows all generators. The theorem also discharges, at the cover level, the "residual conjecture" left open in the docstring of* `Ste.VariablePresheaf` *(gluing holds precisely for the tractable fragments) — provided "tractable fragment" is read as* scope-refinement*, which by Observation 1 below is emphatically* not *computational tractability. Finally, a live instance: the empirical frame corpora of Section 10 are generated by unary constraints, are rectangular (*`isRectangle_feasibilitySet_frame`*), and hence glue over* every *cover (*`frame_cechVanishesCover`*, module* `Ste.FrameRectangular`*) — the completeness of the verified corpus solver is an instance of this theorem's easy direction.*

Remark 2 (No structural conditions on the cover)**.** *Readers who know the CSP decomposition literature will notice something missing: classical local-to-global results (tree-clustering, Vorob'ev-style theorems [21, 23]) demand that the cover's scopes hang together tree-likely (the running intersection property). Theorem 1 imposes nothing of the sort — any cover, any overlap pattern. The reason is the convention of Section 6.1: STE's local sections are* globally extendable by definition*. The classical hypotheses earn their keep when the local tables are merely* locally *consistent rather than genuinely extendable (Section 8.4).*

[For the cohomologist.] *In Abramsky–Brandenburger terms: empirical models are arbitrary compatible presheaf data and can be contextual; the presheaf of genuine sections of a global set cannot be, in the per-section sense — the mechanized* `amb_extension_always` *(Appendix A.2) makes the per-section obstruction identically zero — yet family-level gluing still fails exactly when the cover is too fine for the constraint arity. This is also why no nerve-acyclicity hypothesis appears.*

8.2 Effective outcome 1: gluing decides representability

Theorem 1 is an *iff*, and each direction is an effective statement.

Deciding representability over given scopes. "Can T be written as a conjunction of constraints living inside the scopes $U_1, \ldots, U_m$?" is, by the theorem, the same question as "does every compatible family over U glue?" — and direction (b) even supplies the canonical decomposition when the answer is yes: the section cylinders (`sectionCylinder`), computable from the local section tables alone. At the singleton cover this specializes to a decision procedure for rectangularity (`cechVanishes_iff_rectangular`); at 1 rectangle it meets the representation bounds below (`rectCoverNumber_eq_one_iff_cechVanishes`).

Certifying non-decomposability. Contrapositively, a *single* stuck compatible family over U certifies that no generating family of T refines U — no decomposition of the constraint respects those scopes, so any pipeline treating the scopes as self-contained is unsound for T. The general contrapositive is mechanized (`exists_scope_refiningNone_of_not_cechVanishesCover`, in `Ste.SupportCover`)[1].

8.3 Effective outcome 2: where the hardness lives — gluing width is arity, not treewidth

The gluing-width invariant, stated as an unmechanized sketch in the first draft of this material, is now a Lean definition with its headline example machine-checked (module `Ste.GluingWidth`); only the treewidth-side half of the comparison remains prose, and it is marked below.

Definition 3 (Gluing width; **mechanized**, `GluesAtWidth` in `Ste.GluingWidth`)**.** T glues at width w *(`GluesAtWidth` $T\,w$) if the sheaf condition (`CechVanishesCover`) holds for* T *over some cover all of whose scopes have at most* $w+1$ *variables; the gluing width* $\mathrm{gw}(T)$ *is the least such* w*. By Theorem 1 (`cechVanishesCover_iff_exists_scopedGeneration`),* $\mathrm{gw}(T)$ *is exactly (one less than) the minimal* arity *of a generating constraint family for* T*: the smallest scopes* T *decomposes into. (This last arity reading is an immediate prose consequence of the mechanized iff, not itself a named Lean theorem.)*

Observation 1 (Gluing width is arity, not treewidth — a **two-sided mechanized separation**; vanishing is still not tractability)**.** gw *must not be confused with treewidth, and the separating example is now machine-checked on* both *sides. Gluing side:* $\mathrm{allEqual}(n,\alpha)$ *is generated by the pairwise-equality constraints (`pairEqConstraint`, with scope* $\{i,j\}$ *by `hasSupport_pairEqConstraint`; the generation identity is `allEqual_eq_iInter_pairEqConstraint`), so by Theorem 1(a) its obstruction vanishes over the cover by all pairs (`pairCover`, `cechVanishesCover_allEqual_pairCover`) and it glues at width* 1 *(`allEqual_gluesAtWidth_one`):* $\mathrm{gw}=1$*. Treewidth side — previously a marked sketch, now a theorem (module `Ste.CliqueTreewidth`): the library now has the clique* lower *bound for treewidth (`le_treewidth_of_isClique`, complementing the upper bound `treewidth_le_card` of `Ste.GraphTreewidth`); the pairwise-equality presentation of* allEqual *(`allEqualInstance`) has complete primal graph, so its treewidth is exactly* $n-1$ *(`treewidth_primalGraph_allEqualInstance`; the same statement in bucket-elimination form is `inducedTreewidth_allEqualInstance`). Gluing width* 1 *against treewidth* $n-1$*, both machine-checked: the two invariants are provably different. More brutally: graph colourability is generated by edge-scoped disequality constraints (`properColoring`, `edgeNeqConstraint`), so it Čech-vanishes over the edge cover — now machine-checked (`cechVanishesCover_properColoring_edgeCover`); deciding* $T\neq\emptyset$ *(*k*-colourability,* $k\geq 3$*) is still NP-hard — a classical fact that is* not *mechanized (it is not in Mathlib) and enters this paper only as background.* A vanishing gluing obstruction over the natural cover is compatible with full computational hardness. *All the hardness has been pushed into computing the objects the theorem quantifies over — the genuine local sections* $\mathcal{S}_T(U_j)$ *— and none remains in the gluing.*

[1] Mechanized after the first draft of the underlying exploration memo; the singleton-cover witnesses (`diagonal_not_cechVanishesCover`, `allEqual_not_cechVanishes`) instantiate it: no unary decomposition of these couplings exists.

We regard Observation 1 as the honest answer to "is a vanishing obstruction over a good cover an algorithm?" It is half of one: *gluing is the free half; local sections are the expensive half.* The algorithmic payoff below is precisely the theory of when the expensive half is cheap.

8.4 Effective outcome 3: the algorithmic payoff at bounded width

Bag tables are local sections. The treewidth chain (`Ste.Treewidth` through `Ste.JunctionTree`) is built on the `table` of a constraint on a scope σ: the set of restrictions to σ of solutions. This is *the same construction* as `localSections`, and the identification is now a machine-checked lemma (`table_eq_localSections`, in `Ste.JunctionTree`) — so the junction-tree theorems are, without change and by citation rather than analogy, statements about the section presheaf on the bag cover:

- *Faithfulness: the bag tables jointly remember the constraint.* A scoped constraint is exactly the preimage of its bag table (`HasSupport.eq_preimage_table`), and a global assignment is a solution iff it restricts into every bag table (`mem_joinConstraint_iff_restrict`, `junctionTables_faithful`). This is Theorem 1(b) in concrete clothing: membership in T is decided scope-by-scope.
- *Cost: local sections are small when scopes are small.* A bag of at most $w + 1$ variables over alphabets of size at most k tables in at most k^{w+1} rows (`table_encard_le_pow`); a width-w elimination order materializes at most $n \cdot k^{w+1}$ rows in total (`bucketEliminate_total_space`, `junctionTree_size_le`; linear-in-n form `junctionTree_size_linear` at the minimal duplicate-free width `junctionWidth`, which provably equals the induced width, `junctionWidth_eq_inducedTreewidth`).
- *Computability: bucket elimination computes genuine sections at bounded width.* Projective bucket elimination computes exactly the projection of the solution set onto the remaining variables at each step (`joinConstraint_projectBucketEliminate` and neighbours in `Ste.BucketConsistency`) — genuine local sections, not approximations — decides feasibility (`projectBucketEliminate_decides`, `bucketEliminate_decides`), and extracts a solution backtrack-free (`exists_extension_of_mem_projectBucketEliminate`); the fused statement is `projectBucketEliminate_complete_solver` and, for adaptive consistency proper, `decreasingRun_adaptiveSolver` [18–20].

So the mechanized algorithmic layer already computes the genuine local-section tables over an elimination cover, in $O(n \cdot k^{w+1})$ at induced width w (`inducedTreewidth`, `achievesWidth_inducedTreewidth`; the two-sided bridge to Robertson–Seymour treewidth [31] is `treewidth_primalGraph_le` and `inducedTreewidth_le_treewidth_sub_one`; nonserial dynamic-programming ancestry in 9). Combined with Theorem 1(a) — gluing is free once the generators fit the bags — this is a complete local-to-global pipeline for bounded-width instances: $O(n \cdot k^{w+1})$ preprocessing replaces 2^N enumeration, and bag-local queries read off the section tables soundly by faithfulness.

Consistency algorithms compute pseudo-sections. Local-consistency algorithms [23, 29] compute something weaker than genuine local sections: tables closed under local checks, with no guarantee their entries extend to global solutions. Call these *pseudo-sections.* The gap between pseudo and genuine is exactly where the hardness displaced by Observation 1 ends up living, and the mechanized tractability results are precisely theorems that *pseudo equals genuine* under structural hypotheses: on a chain, arc consistency puts every surviving value on a global solution (`arcConsistent_backtrackFree`, `arcConsistent_chain_solvable`); on a rooted tree, directional arc consistency does the same (`treeArcConsistent_solvable`, `Ste.ConsistencyTree`); at induced width w, directional consistency of the buckets does the same, and bucket elimination establishes it while preserving solutions (`directionalConsistent_solvable` in `Ste.AdaptiveConsistency`; `Ste.BucketConsistency`).

	pseudo-sections (k-consistent tables)	genuine sections (`localSections`)
cheap to compute	yes (local closure)	only at bounded width
entries globally extendable	not guaranteed	always, by definition (cf. `amb_extension_always`)
family gluing	fails generically	iff generators fit the cover (Thm. 1)
where hardness sits	pseudo $\neq$ genuine gap	computing $\mathcal{S}_T$

The conjectural summary (a reading, not a theorem): *the cohomology-shaped content of STE hardness is the pseudo/genuine gap, and every tractability theorem in the library is a statement that the gap closes over a structured cover.*

8.5 Effective outcome 4: representation bounds and certificates

A tempting slogan — "the bigger the obstruction, the bigger the forced representation blow-up" — is settled *negatively*, and machine-checked, in `Ste.RepresentationBounds`. The measuring stick is the *rectangle-cover number* $\rho(T)$ (`rectCoverNumber`): the least number of no-coupling (rectangular) constraints whose *union* is exactly T — "how many coupling-free alternatives describe T?" This is the STE analogue of the nondeterministic communication-complexity cover number [7, 28], with the LP-extension-complexity lineage [22, 35] as outlook. Machine-checked:

- *Bounds.* A *fooling set* — solutions no two of which can be mixed coordinate-wise and stay solutions — forces one rectangle each: $\mathrm{fool}(T) \leq \rho(T) \leq |T|$ (`foolingNumber_le_rectCoverNumber`, `IsFoolingSet.encard_le_of_hasRectCover`, `rectCoverNumber_le_encard`); the mixing engine is `IsRectangle.mix_mem`.
- *The bottom of the scale is the gluing condition*: $\rho(T) = 1 \iff$ the singleton-cover obstruction vanishes (`rectCoverNumber_eq_one_iff_cechVanishes`, `one_lt_rectCoverNumber_iff`).
- *Exact values on the coupling*: $\rho(\mathrm{allEqual}(n, \alpha)) = |\alpha|$ (`rectCoverNumber_allEqual`; fooling set `isFoolingSet_allEqual`, `foolingNumber_allEqual`).
- *No law connects obstruction size to* ρ (`not_forall_cechObstruction_le_rectCoverNumber`, `not_forall_rectCoverNumber_le_cechObstruction`): rectangles give obs $= 0 < 1 = \rho$; $\mathrm{allEqual}(3, \mathrm{Bool})$ gives $\rho = 2 < 6 =$ obs (`rectCoverNumber_lt_cechObstruction_allEqual`); on the two-variable diagonal the two coincide (`diagonal_cechObstruction_eq_rectCoverNumber`) — a coincidence of the smallest example, not a law.

Two representation grammars are in play: ρ measures $\cup$-of-boxes (DNF-like) representations; the junction-tree machinery measures $\cap$-of-cylinders (CNF-over-bags) representations. The machine-checked triple (obs, ρ, width) on allEqual shows all three diverge: obstruction exponential, ρ constant, width maximal.

The remaining direction — stated as an open conjecture in earlier drafts of this material — is now a theorem, mechanized in the module `Ste.ChainFooling`.

Theorem 2 (ρ of the disequality chain; **mechanized**, `Ste.ChainFooling`)**.** *Let* neqChain(n, α) *be the disequality chain on n variables — consecutive coordinates must differ (`neqChain`) — over any alphabet α with three distinct symbols. For chain length $n = 3m$,*

$$2^m \;\le\; \rho\big(\mathrm{neqChain}(3m, \alpha)\big)$$

(`two_pow_le_rectCoverNumber_neqChain`), i.e. $\rho = 2^{\Omega(n)}$. The witness is exactly the "alternating assignments" fooling set the conjecture anticipated: cut the chain into m blocks of three, fill block q with (a, b, c) or (b, a, c) according to a bit vector (`chainWitness`), and the c-separators make any rectangle-mixing of two distinct bit vectors repeat a value across some block boundary — so the 2^m witnesses form a fooling set (`isFoolingSet_range_chainWitness`, via the mixing engine `IsRectangle.mix_mem`). Honest boundary, per the module*: the exact feasible-set count $k(k-1)^{n-1}$ and any matching upper bound on ρ are* not *mechanized; the chain's width — its primal graph is a path, induced width* 1 *— is likewise an inspection-level prose fact, not a named Lean theorem.*

Corollary 1 (Two-sided incomparability of ρ and treewidth — both directions machine-checked)**.** *Neither invariant bounds the other in either direction:* allEqual(n, α) *has constant $\rho = |\alpha|$ (`rectCoverNumber_allEqual`) but maximal treewidth $n - 1$ (`treewidth_primalGraph_allEqualInstance`), while the disequality chain has width* 1 *(a path) but $\rho \ge 2^{n/3}$ (Theorem 2). ρ answers "how many boxes"; width answers "how large a bag"; the mechanization shows each can be minimal while the other is extremal, so neither is a substitute for the other as a compressibility certificate.*

Practitioner's reading: fooling sets are cheap-to-exhibit certificates that no small union-of-boxes representation exists; stuck compatible families over a proposed cover (Section 8.2) are cheap-to-exhibit certificates that no generating family respects those bags. Both certificate kinds are mechanized at the definition level and computed exactly on the coupling family. Cheap NO-certificates steer the expensive machinery: test a candidate bag cover with a small stuck-family search before paying for elimination.

9 Carlson connection points

Does Carlson's cryptanalytic STE [12] map onto the sheaf picture? The honest answer is: *partially, and the partial map is instructive.* We separate three tiers.

9.1 What maps, with mechanized witnesses

Observation accumulation is the constraint axis. Carlson's "more intercepted traffic narrows the key search" (`feasibleKeys_antitone`) is an instance of the axis that glues for free (Section 4, item 1; `partialFeasibilitySet_antitone`, `feasibilitySet_eq_iInter_cover`): observations index property sets, and covers of the *observation* index can never obstruct. The dissertation's accumulation arguments live entirely on this axis, which is why they need no gluing hypotheses.

Residual key counting is fiber counting. Carlson's Lemma 4.1 (`card_consistent_keys`) counts the permutations agreeing with the true key σ on the observed symbol set T: $(|A| - |T|)!$. In the paper's vocabulary, a key's action on T is a partial assignment on the scope T (variables = alphabet symbols, values = alphabet symbols), and the count is the number of *global witnesses of one local section* — the fiber of the restriction map over $\sigma|_T$. Unicity (`Unicity`) is the statement that accumulated observations shrink every such fiber to one point. This reading is faithful to the mechanized statement as it stands, and its formal restatement through the restriction map used by `localSections` is itself mechanized (`card_restrict_fiber`; Section 9.2).

9.2 The scoped-observation theorems (mechanized, `Ste.Carlson.SheafBridge`)

The genuinely sheaf-shaped content of the substitution cipher is this: *an observed ciphertext constrains the key only on the symbols that occur in it* — an observation constraint is a *scoped* constraint in the sense of `HasSupport`, with scope the ciphertext's symbol set (on the decryption-side parametrization of keys). Meanwhile the requirement that the key be a *permutation* — injectivity across all symbols — is a global *coupling*, the cryptanalytic cousin of allEqual's all-different flavor. Both facts, conjectured during the writing of this paper, are now mechanized sorry-free in the main-library module `Ste.Carlson.SheafBridge`: the decryption-side observation constraint `decConstraint` has support exactly the ciphertext's symbol set (`hasSupport_decConstraint`); it is Carlson's `keyPropertySet` of the substitution cipher system `subCipher`, read through key inversion (`mem_keyPropertySet_iff_decConstraint`); the permutation coupling `bijKeys` is *not* rectangular over any nontrivial alphabet (`bijKeys_not_rectangular`); and Carlson's Lemma 4.1 restates through the restriction map used by `localSections` — the permutations restricting on T to a given local section number $(|A| - |T|)!$ (`card_restrict_fiber`, derived from `card_consistent_keys`). With these, the fiber-counting reading of Section 9 is a theorem, not an analogy.

So Carlson's instance factors exactly along this paper's main axis: scoped evidence (gluable, by Theorem 1(a), over any cover the symbol sets refine) against one global coupling (the permutation constraint, which no proper cover swallows). And the factorization places the two applied instantiations of this paper at *opposite ends of the obstruction spectrum*, machine-checked at both ends: the frame corpora of Section 10 are rectangular — obstruction zero over every cover (`isRectangle_feasibilitySet_frame`, `frame_cechVanishesCover`) — which is why a linear per-variable solver decides them completely, while Carlson's substitution-cipher keys are provably *non*-rectangular (`bijKeys_not_rectangular`) — genuinely coupled, on the nontrivial side of the spectrum, which is why cryptanalysis cannot be finished by per-symbol reasoning and Shannon-style accumulation toward unicity is needed at all.

9.3 What does not map

Two honest negatives. First, Carlson's hypothesis space is a key space, not a product of variables; the sheaf reading above requires re-embedding keys into the function space $A \to A$ and carrying the permutation constraint separately. Nothing in the dissertation uses cover-locality on the key's coordinates; the productive structure there is *class-level reduction* — embedding one cipher class's keys in another's (`permutation_reduces_to_substitution`, `coperm_injective`, `substitution_not_reducible_to_permutation`, `psp_reduces_to_substitution`) — which is group theory, not gluing, and we see no gluing-theoretic content in it. Second, the dissertation's information-theoretic layer (unicity distances, entropy of meaningful text) has no mechanized counterpart in this project and no sheaf reading we would defend; connecting the obstruction counts of Section 8.5 to unicity-distance estimates is *speculation* and is recorded as such in Section 12.

10 Empirical analysis and constructive proof-of-work

The project's empirical layer runs a *verified* STE solver on frame corpora extracted from historical texts. This section reports what ran, exactly what the solver computes, and what the cohomological theory adds — with the pipeline's trust boundary stated plainly.

10.1 The frame-corpus constraint class and the verified solver

An empirical corpus is a *frame* family: each author a fixes a partial assignment frame $a : V \to \text{Option}\, W$ of values to variables (silence = `none`), and the property set of a is the set of total assignments agreeing with a wherever a speaks (`frameConstraint`, in `Ste.FiniteInstance`). For this class the feasibility question collapses to a decidable local check, and the collapse is machine-checked twice over: the corpus is jointly satisfiable iff no two authors assign the same variable different values (`Consistent`, with the bridge `nonempty_iff_consistent` and the verdict form `feasibilitySet_eq_empty_iff`); and pairwise consistency is in turn equivalent to a *per-variable* condition — every variable's disagreement degree (its count of distinct asserted values, `disagreementDegree`, characterized by `agree_iff_degree_le_one`) is at most one (`consistent_iff_forall_disagreementDegree_le_one`).

The executable (`lean/TidesExe.lean`) parses a `frames.json` at runtime and computes the verdict as exactly that last, cheapest form: a single per-variable scan of the disagreement degrees, sound by `consistent_iff_forall_disagreementDegree_le_one`. **To be precise about what runs: the solver's algorithm is the per-variable degree scan of `Ste.FiniteInstance`; it imports no cohomology module, and no Čech computation executes on any corpus.** The trust boundary is the JSON parser and Lean's evaluator; the aggregation logic is the verified one. What the cohomology contributes is not the solver's code but the theorem that this per-variable algorithm is *complete* for the constraint class — Section 10.3.

10.2 The runs

Three corpora, one pipeline (LLM extraction of frames from period texts; the verdicts computed by the verified layer):

- *Tides* (`Ste.TidesCorpus`, `Ste.TidesComputable`): pre-Newtonian authors on the cause of the tides, canonicalized to a 7-variable schema. Verdict: globally *inconsistent*, and pervasively so — the five-author "attraction" camp is internally consistent on the moon's role but not overall (`tides_inconsistent`, `attraction_camp_moon_consistent`); the kernel-computed verdict (`feasible_eq_empty_computed`) agrees with the hand-proved one, and the corpus is definitionally an instance of the general frame class (`constraint_eq_frameConstraint`).
- *Federalist* (86 essays; 27): 10 variables; inconsistent but *narrowly* — 7 of 10 variables unanimous where spoken, exactly 3 conflict sites carrying the whole obstruction.
- *Ratification, two camps* (the 86 Federalist essays plus 73 Anti-Federalist essays; 159 documents): the consistent core shatters — all 10 variables become conflict sites. The contrast $3 \to 10$ is the sharp empirical signature of adding an opposing camp.

10.3 What the cohomology adds: why the cheap check is the right one

Here is the division of labor, now closed end-to-end. The solver runs a *per-variable* check. The theory of this paper proves *why per-variable is enough for this constraint class* — why no higher-order gluing failure can hide in a frame corpus — and the proof is machine-checked in the module `Ste.FrameRectangular`.

The plain argument, now theorem by theorem: each author's `frameConstraint` is a rectangle — a product of per-variable allowed-value sets ($\{x\}$ where the author asserts x, everything where silent) — machine-checked as `frameConstraint_isRectangle`; the corpus feasible set, an intersection of rectangles, is itself a rectangle (`isRectangle_feasibilitySet_frame`); hence, by the sheaf theorem for rectangles (`rectangular_cechVanishesCover`, Section 7), the feasible set of a frame corpus Čech-vanishes over *every* cover of the variables (`frame_cechVanishesCover`): **zero coupling obstruction, for any way of localizing whatsoever**. There are no phantoms: locally consistent readings of a frame corpus are always globally realizable. That is exactly why the decidable surrogate `Consistent` coincides with genuine feasibility (`nonempty_iff_consistent`) and why even the coarser per-variable degree scan the executable actually runs decides it (`consistent_iff_forall_disagreementDegree_le_one`): the corpora sit at the trivial end of the obstruction spectrum of Section 6.2 — rectangular, obstruction 0, the $\rho = 1$ regime of Section 8.5 — which is precisely the regime where a linear-time local solver is a *complete* gluing test rather than a heuristic. And the same theory says exactly where this solver would stop being complete: the moment a constraint couples variables (a non-rectangular `frameConstraint` cannot exist, but relational claims would leave the class), the diagonal of Section 6.2 is the smallest counterexample. Section 11 restates the same rectangularity fact in a third, cover-free language — frame corpora have Helly number 2 (`frame_hasHelly_two`) — and the co-citation with `frame_cechVanishesCover` is itself a single mechanized statement (`frame_hasHelly_two_and_cechVanishesCover`).

Two consequences worth stating plainly. First, the empirical runs are a genuine, if structurally easy, instance of the paper's subject: a local-reasoning solver whose completeness is exactly a gluing fact. Second, the interesting failure mode — coupled constraints, where pairwise reasoning is *not* complete (Section 6.2) — does not arise for frame corpora, but will arise the moment extraction produces *relational* claims ("v_1 and v_2 stand in relation R") rather than variable-value assignments. A solver for that regime, carrying its gluing certificate explicitly (compute bag-local sections, check compatibility, invoke Theorem 1 for soundness), is the natural constructive next step; **no such cohomology-labelled solver exists yet**, and we flag it as future work rather than claim it.

11 Metric-free discriminators: what feasible families have that infeasible ones lack

Behind the whole development sits a blunt question: *what does a structure of subsets that is feasible have that a non-feasible one does not?* Combettes' analytic toolkit answers with metric structure — distances, projections, convexity in Hilbert space [16] — but nothing in Sections 2–10 carries a useful metric, and the covers and obstructions of this paper were built deliberately without one. This section records two further answers, both newly mechanized sorry-free in the main library, and both strictly lighter-weight than a metric space: a *combinatorial* discriminator — a Helly number, with its closure system (module `Ste.HellyConsistency`) — and an *order-theoretic* one — a submodular rank on the partition lattice (module `Ste.PartitionRank`). The first needs only intersections of property sets; the second only counts blocks of partitions. Both detect exactly the rectangularity phenomenon that Sections 6.2 and 10.3 located cohomologically, and the coincidence is itself a theorem (`frame_hasHelly_two_and_cechVanishesCover`, below).

11.1 Discriminator 1: the Helly number — when is feasibility a 2-local property?

Say a family $(S_i)_{i \in I}$ of property sets is *k-wise consistent* (`KWiseConsistent`) if every subfamily of at most k of them has nonempty intersection, and that it *has the Helly property at level k* (`HasHelly`) if k-wise consistency already forces the full feasibility set $T = \bigcap_i S_i$ to be nonempty. (The names honor Helly's classical theorem for convex sets in $\mathbb{R}^d$ — every $(d+1)$-wise intersecting finite family of convex sets is globally intersecting — which is the metric-laden ancestor of the property; here no convexity, norm, or dimension appears.) One direction is free and holds for every family: a nonempty feasibility set witnesses every finite subfamily's nonemptiness (`kWiseConsistent_of_feasibilitySet_nonempty`, `hasHelly_of_feasibilitySet_nonempty`; the bridge between Finset-indexed intersections and `partialFeasibilitySet` is `partialFeasibilitySet_coe_finset`, and k-wise consistency is antitone in k, `KWiseConsistent.mono`). The discriminating content is the converse, and it splits the world exactly along the rectangularity line.

Theorem 3 (Frame corpora have Helly number 2; **mechanized**, `frame_hasHelly_two` in `Ste.HellyConsistency`)**.** *For every frame corpus (property sets* `frameConstraint`*, Section 10), pairwise consistency implies global consistency: if every pair of authors is jointly satisfiable, the whole corpus is. Feasibility of a frame family is an entirely* 2*-local property — the complete feasibility test inspects pairs and nothing larger.*

Proposition 1 (The general subset lattice is not Helly-2; **mechanized**, `exists_not_hasHelly_two`)**.** *There is a family of three subsets of a three-element space —* $S_0 = \{0, 1\}$*,* $S_1 = \{1, 2\}$*,* $S_2 = \{0, 2\}$ *— that is pairwise consistent (every two of the sets meet) yet globally infeasible: the triple intersection is empty.*

Together, the two results are the discriminator in its sharpest form. An infeasible family can *look* feasible to every bounded-size probe: the triangle of Proposition 1 passes every pairwise test and still has no solution — the consistency-number analogue of the phantoms of Section 6.2. What the frame class has, and general families lack, is the theorem that bounded-size probes are conclusive. "Helly number 2" is a property a class of subset structures either has or does not have, stated entirely in the language of intersections — and rectangularity is exactly what buys it.

The closure system. The same module records the order-theoretic backbone under the discriminator: the operator $\mathrm{cl}(X) = \bigcap \{S_i \mid X \subseteq S_i\}$ (`cl`) — the smallest constraint-expressible set containing X — is a genuine closure operator, machine-checked as extensive (`cl_extensive`), monotone (`cl_monotone`), and idempotent (`cl_idempotent`); its closed sets (`IsClClosed`) are stable under arbitrary intersection (`isClClosed_iInter`), so they form a Moore family. Every property-set family thus carries a canonical lattice of *expressible* sets, with feasibility a question about the bottom of that lattice — structure that exists with no metric in sight.

Same phenomenon, two languages. The Helly bound and the vanishing Čech obstruction of Section 10.3 are not merely analogous: both are downstream of the single structural fact that the frame feasible set is a rectangle (`isRectangle_feasibilitySet_frame`), and their conjunction is one mechanized statement, `frame_hasHelly_two_and_cechVanishesCover`: every frame corpus simultaneously has Helly number 2 and zero gluing obstruction over every cover of the variables (`frame_cechVanishesCover`). The combinatorial probe (intersect small subfamilies) and the topological probe (try to glue local sections) are two instruments pointed at the same absence of coupling.

11.2 Discriminator 2: Shannon's information lattice — the submodular partition rank

The second discriminator lives on the corpus axis of Section 4 (item 3) and makes precise the sense in which a feasible corpus *carries information* with no metric, probability, or entropy in the formalism. The reading is Shannon's: his 1953 "lattice theory of information" models an information source as a partition of a fixed space, orders sources by refinement, and reads entropy as a rank function on the resulting lattice, submodular in the shape $H(A \vee B) + H(A \wedge B) \le H(A) + H(B)$. The mechanization keeps the lattice and replaces entropy by counting.

In the dynamic-frame model (`Ste.DynamicFrame`), a *reading* of the corpus is an equivalence relation on claims — which claims name the same underlying frame — and each feasible normalization hypothesis h induces one, packaged as an element `frameSetoid` h of Mathlib's complete lattice of setoids (namespace `STE.DynamicFrame.Model`, module `Ste.PartitionRank`). Readings are ordered by refinement, and the corpus's *forced-identity* (must-coreference) relation sits at a distinguished lattice position:

Theorem 4 (Must-coreference is the lattice meet; **mechanized**, `mustSetoid_eq_sInf`)**.** *For any document set D, the must-coreference setoid of D equals* sInf *of the frame setoids of the feasible hypotheses: two claims are forced to corefer exactly when* every *surviving hypothesis identifies them. Feasibility, read on the partition lattice, is a meet — the common refinement of the surviving readings. In particular the must-partition refines every feasible reading, and grows monotonically with evidence: adding documents only adds forced identifications (`rank_mustSetoid_mono` below is its rank shadow).*

The invariant the feasible family carries is the *rank* of these partitions (namespace `STE.PartitionRank`): for a finite carrier, $r(P) = |\text{carrier}| - \#\text{blocks}(P)$ (`blockCount`, `rank`) — the number of merges P has committed to. Block counting behaves as the lattice demands: every partition has at least one block (`one_le_blockCount`), coarsening only lowers the count (`blockCount_antitone`), and the covering step is exact — merging one previously-unrelated pair (`mergePair`) drops the block count by exactly one (`blockCount_mergePair`), the counting engine that makes the partition lattice geometric rather than merely ordered. On that engine the module proves the Shannon-shaped law:

Theorem 5 (The partition rank is submodular; **mechanized**, `rank_submodular` in `Ste.PartitionRank`)**.** *For all partitions A, B of a finite carrier,*

$$r(A \sqcup B) + r(A \sqcap B) \ \le\ r(A) + r(B),$$

equivalently the block count is supermodular (`blockCount_supermodular`). This is Shannon's information-lattice inequality with counting rank in place of entropy — the metric-free shadow of entropy submodularity.

Transferred back to STE (namespace `STE.DynamicFrame.Model`), the feasible corpus hands us a canonical, machine-checked invariant package: the consensus never claims more forced-merge information than any single surviving reading (`rank_mustSetoid_le_frameSetoid`); coreference information is monotone in the corpus (`rank_mustSetoid_mono`); and the must-partitions of two corpora obey the submodular law (`rank_mustSetoid_submodular`). This is the promised "lighter than a metric space" structure in full: a refinement order, a meet, and a rank obeying the entropy inequality — nothing else. A feasible corpus *has* a well-behaved information content in Shannon's lattice sense; the three inequalities are what having it means, and all three are theorems.

Honest boundary. The genuine entropy layer — $H = \log|\text{feasible}|$ on hypothesis space, the map relating hypothesis-space and claim-space partitions, and the conjectured dichotomy that rank decomposes additively over rectangular families while genuine coupling makes joins strictly superadditive — is recorded as an explicit conjecture in the module's source and is *not* mechanized; Section 12 lists it as such.

12 Conclusion and future work

12.1 What is machine-checked, in one paragraph

The constraint side of STE glues for free (`feasibilitySet_eq_iInter_cover`); the corpus side glues with an exact computable correction (`mem_feasible_union_iff`); on the variable side, per-variable gluing characterizes the absence of coupling exactly (`cechVanishes_iff_rectangular`), coupling-free constraints glue over every cover with unique glues (`rectangular_glueCover_existsUnique`), couplings genuinely fail (`diagonal_not_cechVanishesCover`) with exactly computed, exponentially growing phantom counts (`cechObstruction_allEqual`, `cechObstruction_allEqual_ge`), and the boundary is exact: the support–cover gluing theorem (Theorem 1, `Ste.SupportCover`) with its certificate contrapositive (`exists_scope_refiningNone_of_not_cechVanishesCover`). The gluing-width invariant is a Lean definition with its headline computed two-sidedly: allEqual glues at width 1 (`GluesAtWidth`, `allEqual_gluesAtWidth_one`, via `cechVanishesCover_allEqual_pairCover`) while its treewidth is $n - 1$ (`treewidth_primalGraph_allEqualInstance`, via the clique lower bound `le_treewidth_of_isClique`), and graph colourability Čech-vanishes over the edge cover (`cechVanishesCover_properColoring_edgeCover`). Cover-shape-only invariants are provably blind (`cechH1_subsingleton`, Appendix A.1) while the section-aware linear obstruction fires over every nontrivial ring (`mixedFamily_not_linearlyGlues`) even though the per-section AMB obstruction cannot (`amb_extension_always`). The rectangle-cover number is sandwiched, characterized at 1 by gluing, computed on the coupling, provably incomparable to the obstruction magnitude (`not_forall_cechObstruction_le_rectCoverNumber` and mirror), and provably incomparable to treewidth in both directions (Corollary 1: `rectCoverNumber_allEqual` against `treewidth_primalGraph_allEqualInstance`, and `two_pow_le_rectCoverNumber_neqChain` on the width-1 chain). The bounded-width stack — tables, bucket elimination, junction trees, adaptive consistency — is mechanized with faithfulness, $O(n \cdot k^{w+1})$ size, a fused complete-solver statement (`projectBucketEliminate_complete_solver`), and the identifications `table_eq_localSections` and `junctionWidth_eq_inducedTreewidth` that make it literally a theory of the section presheaf on the bag cover. Carlson's STE recast, counting lemma, and reduction chain are mechanized (`Ste.Carlson`), and so is the sheaf bridge: observation constraints are scoped (`hasSupport_decConstraint`), residual key counting is fiber counting (`card_restrict_fiber`), and the permutation coupling is non-rectangular (`bijKeys_not_rectangular`). The frame-corpus solver equivalence is mechanized down to the per-variable scan the executable runs (`nonempty_iff_consistent`, `consistent_iff_forall_disagreementDegree_le_one`), it ran on three corpora, and its completeness is itself a theorem: frame corpora are rectangular with zero obstruction over every cover (`isRectangle_feasibilitySet_frame`, `frame_cechVanishesCover`). The same rectangularity is captured by two further metric-free discriminators (Section 11): frame corpora have Helly number 2 while a three-set family witnesses general failure (`frame_hasHelly_two`, `exists_not_hasHelly_two`, jointly with Čech vanishing as `frame_hasHelly_two_and_cechVanishesCover`), the property sets generate a mechanized closure system whose closed sets form a Moore family (`cl_extensive`, `cl_monotone`, `cl_idempotent`, `isClClosed_iInter`); and on the corpus axis the forced-coreference partition is the lattice meet of the feasible readings (`mustSetoid_eq_sInf`), with a partition rank that is exactly computed on covering steps (`blockCount_mergePair`), submodular (`blockCount_supermodular`, `rank_submodular`), bounded by every surviving reading (`rank_mustSetoid_le_frameSetoid`), monotone in evidence (`rank_mustSetoid_mono`), and submodular on must-partitions (`rank_mustSetoid_submodular`).

12.2 What remains conjectural

The prose arity reading of gw in Definition 3; NP-hardness of colourability (cited, not mechanized); around Theorem 2, the chain's exact feasible-set count $k(k-1)^{n-1}$, any matching upper bound on ρ, and the chain's width-1 fact as a named Lean theorem; a general lower bound against width-w schemes (the junction-tree file's outlook item, for which `cechObstruction_allEqual_ge` is the margin-cover prototype); the pseudo/genuine-gap reading of tractability (Section 8.4); the bag-local linear relaxation with exactness controlled by the twisted invariant (Appendix A.2); the entropy layer over the partition rank — $H = \log|\text{feasible}|$, the hypothesis-to-claim partition map, and the additive-versus-coupled rank dichotomy (Section 11.2) — stated as a conjecture in the source of `Ste.PartitionRank`; and any connection between obstruction counts and unicity distance (Section 9) — explicitly speculation.

12.3 Proof obligations: dispatched and discharged

The exact obligations below were dispatched to Lean mechanization efforts during the writing of this paper. All three batches have since been discharged *and merged to the main library*, and the relevant sections cite them as ordinary main-library theorems; the record is kept here for provenance, as a worked example of the bright-line workflow (conjecture → dispatched → discharged → merged, with the paper's wording tracking each transition).

Proof Obligation 1 (Carlson sheaf bridge — **discharged and merged**). *(a) The decryption-side observation constraint of the substitution cipher has `HasSupport` equal to the symbol set of the ciphertext. (b) Its equivalence with `keyPropertySet` of the substitution `CipherSystem` via key inversion. (c) The permutation coupling $\{g : A \to A \mid g \text{ bijective}\}$ is not rectangular over a nontrivial alphabet. (d) Carlson's Lemma 4.1 restated through the restriction map (from `card_consistent_keys`).* Status: mechanized and merged to the main library as module `Ste.Carlson.SheafBridge` (`hasSupport_decConstraint`, `mem_keyPropertySet_iff_decConstraint`, `bijKeys_not_rectangular`, `card_restrict_fiber`); Section 9.2 cites them as main-library theorems.

Proof Obligation 2 (Frame rectangularity — **discharged and merged**). *(a) `IsRectangle` of each author's `frameConstraint`; (b) `IsRectangle` of the corpus feasible set; (c) hence `CechVanishesCover` for every cover of the variables.* Status: mechanized and merged to the main library as module `Ste.FrameRectangular` (`frameConstraint_isRectangle`, `isRectangle_feasibilitySet_frame`, `frame_cechVanishesCover`); Section 10.3 now cites these as main-library theorems.

Proof Obligation 3 (Near-term backlog — **discharged and merged**). *The `table` = `localSections` identification; a pair-cover gluing statement for* allEqual *and a Lean-level gluing-width definition.* Status: mechanized and merged (`table_eq_localSections` and `junctionWidth_eq_inducedTreewidth` in `Ste.JunctionTree`; `GluesAtWidth`, `cechVanishesCover_allEqual_pairCover`, `allEqual_gluesAtWidth_one`, and `cechVanishesCover_properColoring_edgeCover` in `Ste.GluingWidth`); Sections 8.3 and 8.4 now cite them as main-library theorems.

12.4 Further directions

A pseudo-section presheaf as a first-class object, with the unifying vanishing theorem "running-intersection cover + directional consistency ⇒ pseudo = genuine" subsuming the chain, tree, and adaptive cases; the variable-side analogue of the corpus axis' exact correction term (splice two partial solutions, collect straddling constraints), then its k-piece iteration; a bag-local must/may/cannot query theorem over junction tables; the twisted degree-one theory (Appendix A.2) toward exactness guarantees for linear relaxations in the style of Abramsky et al. [6]; and the relational-claims solver of Section 10 — the first solver that would carry its gluing certificate explicitly. Pleasingly for a paper about cohomology, the content of its main theorem is that in set-theoretic estimation the cohomology, properly aimed, is exactly the bookkeeping of who is allowed to talk to whom — and that bookkeeping is machine-checked.

A For the cohomologist: what the coefficients see

This appendix is the paper's heavy-machinery annex. Nothing in the main text depends on it beyond the two plain-language morals already stated there: cover-shape-only invariants are provably blind (A.1), and a section-aware linear invariant fires one degree lower than the AMB program would suggest (A.2). The Abramsky–Brandenburger program grades gluing failure cohomologically by linearizing the section presheaf [3, 4]; the library mechanizes two coefficient regimes, and the contrast between them is, in our view, the single most instructive machine-checked fact in this layer.

A.1 Constant coefficients see nothing — a machine-checked no-free-lunch

`Ste.CechComplex` builds the constant-coefficient cochain complex on the full nerve of a cover index — modules $C^0 = \iota \to M$, $C^1 = \iota \to \iota \to M$, ... with the standard alternating coboundaries (`cechD0`, `cechD1`, `cechD2`), the complex identities (`cechComplex_d_comp_d`, `cechComplex_d2_comp_d1`), and honest quotient modules $\check{H}^1, \check{H}^2$ (`cechH1`, `cechH2`). The vanishing theorem (`cechH1_subsingleton`, `cechH2_subsingleton`) proves $\check{H}^1 = \check{H}^2 = 0$ over any nonempty index type: the full nerve is a simplex, and a simplex is acyclic; $\check{H}^0$ is exactly the coefficients (`cechH0EquivCoeff`).

The computational moral: *any invariant computed from the combinatorics of the cover alone — ignoring the section data — is useless for detecting coupling*; the machine checks that it is identically zero. Whatever a "cohomological speedup" means, its input must include the tables of local sections — exactly the data the CSP algorithms of Section 8.4 already manipulate. This is consistent with the contextuality literature, where the interesting classes live in relative cohomology with *presheaf* coefficients [4, 14].

A.2 Twisted coefficients see the coupling — one degree lower than expected

`Ste.TwistedCech` builds the honest twisted complex: coefficients in the linearized section presheaf $F_R(W) = (\mathcal{S}_T(W)) \to_f R$ (free R-module on genuine local sections, `twistedCoeff`), restriction by pushforward along section restriction (`sectionRes`, `twistedRes`, functorial by `twistedRes_twistedRes`), coboundaries `twistedD0`, `twistedD1` with $d^1 \circ d^0 = 0$ (`twisted_d1_comp_d0`), and $\check{H}^0$ as `twistedH0`. Two machine-checked results locate the STE coupling precisely.

1. *The AMB per-section obstruction degenerates in STE* (`amb_extension_always`). Abramsky–Mansfield–Barbosa attach to a single local section s_1 a class $\gamma(s_1) \in \check{H}^1$ vanishing iff s_1 extends to a compatible family of the linearized presheaf [4, Prop. 4.2]. The mechanization proves that in STE this extension condition holds for *every* section of *every* constraint over *every* cover — because `localSections` only ever contains restrictions of global solutions, the extension witness is free. The $\check{H}^1$-graded per-section obstruction is structurally blind here:

STE's section presheaf bakes global extendability into its stalks. (The long exact sequence and the literal connecting map γ are *not* mechanized; what is proven is the extension condition AMB prove equivalent to $\gamma = 0$.)

2. *The obstruction that fires lives in the cokernel of* $F_R(V) \to \check{H}^0(U, F_R)$ (`mixedFamily_not_linearlyGlues`, `diagonal_mixed_cocycle_not_in_globalRange`). A compatible family's point-mass cochain is a twisted 0-cocycle (`familyCochain_mem_twistedH0`); the family *glues R-linearly* (`LinearlyGlues`) iff that cocycle lifts to a formal R-combination of global solutions (`linearlyGlues_iff_mem_range`). For the two-bit coupling and its singleton cover, the mixed family fails to glue linearly over *every nontrivial commutative ring* — negative and rational coefficients included. This is strictly stronger than the set-level failure (`diagonal_gluing_fails`): in the AMB world every no-signalling model has a global section over the signed reals [3], but the STE coupling is not even signed-linearly explainable by global data. Set-level gluing always implies linear gluing (`GluesCover.linearlyGlues`), and rectangles glue linearly over every cover (`rectangular_linearlyGlues`), so nonmembership is a genuine obstruction invariant (`diagonal_twisted_obstruction_detects`).

Whether the twisted $\check{H}^1$ (carried as the submodule pair `twistedCoboundaries1` $\leq$ ker `twistedD1`, with triviality expressed as `TwistedH1Trivial`; the literal quotient type is blocked by a typeclass diamond, documented in the file) vanishes for pairwise-disjoint covers is stated as *open* in the source and remains open here. It is the first stepping stone toward exactness guarantees for bag-local linear relaxations — signed/affine analogues of the k-consistency hierarchy, cousins of the contextual-fraction linear programs of Abramsky et al. [6] — with the standing caution from the literature [1, 14, 30] that invariants of this style can have false negatives; the mechanized `amb_extension_always` is precisely such a false-negative theorem.

[For the cohomologist.] On the corpus axis the presheaf is separated (identical ambient hypothesis on overlaps), so its theory closes at $\check{H}^0$; a Mayer–Vietoris organization of iterated corpus splits (standard machinery, e.g. 11) would carry correction terms that are precisely the `crossObstruction` sets, and separatedness makes us expect immediate degeneration — a polite way of saying the corpus axis needs no cohomology beyond what is mechanized. A bi-indexed presheaf on (corpora) $\times$ (scopes), corpus direction separated, variable direction carrying the twisted theory, is (frankly speculative) outlook.

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